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A DISCUSSIONS

Applications. StreamingLLM is particularly suited for streaming applications, such as multi-round dialogues, where continuous operation without heavy reliance on extensive memory or historical data is crucial. For instance, in a daily assistant application based on LLMs, StreamingLLM enables the model to function seamlessly over extended periods. It bases its responses on recent interactions, thus avoiding the need for frequent cache refreshes. Traditional methods might require resetting the cache when the conversation length surpasses the training length, leading to a loss of recent context, or they might need to recompute key-value (KV) states from recent text history, which can be inefficient.

Limitations. While StreamingLLM improves the efficiency of LLMs in streaming contexts, it does not extend the models’ context window or enhance their long-term memory capabilities. As detailed in Section C, the model is limited to operating within the confines of its current cache. Consequently, StreamingLLM is not suitable for tasks that demand long-term memory and extensive data dependency, such as long document question-answering (QA) and summarization. However, it excels in scenarios only requiring short-term memory, like daily conversations and short document QA, where its strength lies in generating coherent text from recent context without the need for cache refreshment.

Broader Societal Impacts. StreamingLLM significantly enhances the efficiency and accessibility of LLMs, democratizing their use across various sectors. By enabling nonstop and rapid interactions in applications like conversational agents, StreamingLLM improves user experiences, especially in scenarios requiring fixed-length models. This advancement allows for more seamless and contextually aware dialogues, potentially benefiting sectors like education, healthcare, and customer service. Additionally, StreamingLLM’s efficiency in processing reduces the computational load, aligning with the need for environmentally sustainable AI technologies. This aspect is crucial in making advanced AI tools more accessible in regions with limited technological resources. However, the potential negative impacts of StreamingLLM mirror those associated with general language models, such as misinformation and biased content generation risks. It’s essential to address these risks with robust ethical guidelines and safeguards. In summary, while StreamingLLM shares some risks common to language models, its positive contributions towards enhancing user experience, democratizing AI access, and promoting sustainability are noteworthy. These benefits underscore the importance of responsible deployment and ethical use of this technology.

B ADDITIONAL RELATED WORKS

Sparse Transformers. The literature on efficient Transformer models primarily focuses on reducing the computational and memory complexity of the self-attention mechanism. A relevant line of work involves sparsifying the attention matrix by restricting the field of view to fixed, predefined patterns, such as local windows or block patterns with fixed strides (Tay et al., 2022). Sparse Transformer (Child et al., 2019) introduces sparse factorizations of the attention matrix, reducing the computational complexity of attention to $O(n\sqrt{n})$. LongFormer (Beltagy et al., 2020) combines dilated local windowed attention with task-motivated global attention. Extended Transformer Construction (ETC) Ainslie et al. (2020) presents a novel global-local attention mechanism, incorporating four types of attention patterns: global-to-global, local-to-local, local-to-global, and global-to-local. Building on ETC, BigBird (Zaheer et al., 2020a) proposes another linear complexity attention alternative, utilizing global tokens, local sliding window attentions, and random attention. However, these methods have several limitations. First, Sparse Transformer and ETC require custom GPU kernels for a specific block-sparse variant of matrix-matrix multiplication. Second, LongFormer, ETC, and BigBird all rely on a global attention pattern, which is unsuitable for autoregressive language models. Third, these methods are incompatible with pre-trained models, necessitating retraining from scratch. In contrast, our method offers ease of implementation using standard GPU kernels and is compatible with pre-trained autoregressive language models using dense attention, which are prevalent in the NLP community. This compatibility provides a significant advantage, allowing for the leveraging of existing pre-trained models without any fine-tuning.

Concurrent Works. Our research coincides with the work of Han et al., who conducted a theoretical study on the length generalization failure of language models, identifying three out-of-distribution factors. Their approach, inspired by this analysis, involves employing a “Λ”-shaped attention pattern

Table 7: Accuracy (in %) on StreamEval with increasing query-answer distance. Each line in StreamEval contains 23 tokens. Accuracies are averaged over 100 samples, and each sample contains 100 queries.

Llama-2-7B-32K-Instruct		Cache Config			
Line Distances	Token Distances	4+2044	4+4092	4+8188	4+16380
20	460	85.80	84.60	81.15	77.65
40	920	80.35	83.80	81.25	77.50
60	1380	79.15	82.80	81.50	78.50
80	1840	75.30	77.15	76.40	73.80
100	2300	0.00	61.60	50.10	40.50
150	3450	0.00	68.20	58.30	38.45
200	4600	0.00	0.00	62.75	46.90
400	9200	0.00	0.00	0.00	45.70
600	13800	0.00	0.00	0.00	28.50
800	18400	0.00	0.00	0.00	0.00
1000	23000	0.00	0.00	0.00	0.00

and reconfiguring position encoding distances to enhance length generalization in LLMs. This approach bears a resemblance to our methodology. However, our work uncovers the “attention sink” phenomenon, wherein Transformer models tend to assign high attention scores to initial tokens with small semantics. This phenomenon extends beyond the scope of length generalization failure, indicating a more pervasive issue in Transformer models. We observe this “attention sink” behavior not only in auto-regressive language models but also in encoder Transformers such as BERT (see Section H), and Vision Transformers (ViTs) (Darcet et al., 2023), suggesting its broader prevalence in Transformer architectures. To mitigate the “attention sink” phenomenon, we propose the introduction of a learnable sink token during pre-training, and we support our findings with extensive ablation studies.

In parallel, Darcet et al. observed similar attention concentration on random background patch tokens in Vision Transformers, termed as “registers.” These registers act as repositories for global image information. Their solution, adding dedicated “register” tokens, aims to balance attention distribution. Our finding of “attention sinks” parallels this concept. In our paper, the “attention sinks” are initial tokens that disproportionately attract attention from subsequent tokens. Introducing a dedicated sink token during pre-training prevents the model from inappropriately using content tokens as attention sinks, leading to more effective attention distribution. However, a key difference exists: “registers” in Vision Transformers function as global information holders within intermediate layers, whereas our “attention sinks” are positioned as initial tokens in autoregressive models. This positional variance suggests that the softmax function in attention computation might play a more fundamental role in the emergence of attention sinks.

C ACCURACY ON STREAMEVAL WITH INCREASING QUERY-ANSWER LINE DISTANCE

To assess StreamingLLM’s handling of extended inputs, we evaluated the Llama-2-7B-32K-Instruct model on StreamEval, focusing on different query-answer line distances under various cache configurations. In StreamEval, each line consists of 23 tokens, making the line distances equivalent to token distances of $23 \times \text{line distances}$. Accuracy was calculated by averaging results over 100 samples, with each sample comprising 100 queries. Table 7 illustrates that StreamingLLM retains accuracy when the token distance between the query and answer is within the cache size. However, accuracy diminishes as this distance increases and eventually drops to zero when it surpasses the cache capacity.

These results demonstrate that while StreamingLLM is effective in generating coherent text based on recent context, it cannot extend the context length of language models. These results also emphasize a broader challenge in current language models: their inability to fully utilize context information within the cache, a finding that aligns with the observations made by Liu et al..